CASTELNUOVO-MUMFORD REGULARITY IS PRESERVED UNDER EXTENSION OF THE GROUND FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Flat base change for projective space over a field extension. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology. The identification of cohomology groups.

Source reference. Original reference 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k'/k be an extension of fields. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$. Let $\mathcal{F}'$ be the pullback of $\mathcal{F}$ to $\mathbf{P}^n_{k'}$. Then $\mathcal{F}$ is m-regular if and only if $\mathcal{F}'$ is m-regular.*

Proof. This is true because

$$H^i(\mathbf{P}^n_{k'}, \mathcal{F}') = H^i(\mathbf{P}^n_k, \mathcal{F}) \otimes_k k'$$

by flat base change, see Cohomology of Schemes, Lemma 02KH. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.